

# Black-Box Probing for Engineered Dormant Backdoors

## Executive summary

Engineered dormant backdoors are usually defined as models that behave normally on clean inputs yet switch into a distinct behavior regime when a trigger pattern is present—classically demonstrated in **BadNets** and extended widely to NLP/LLMs. <sup>1</sup>

With only black-box access, the most reliable approach is a **factorial probe design** (base prompts × perturbations), followed by **embedding-based drift + clustering + distributional two-sample tests**, then **minimality/robustness** validation and strict **false-positive control**. <sup>2</sup>

Language-like triggers are plausible in engineered settings because backdoors can be keyed by **rare word combinations** (“combination locks”) or even hidden **linguistic-style** features, not just weird symbols. <sup>3</sup>

## Mechanistic framing for language-like triggers

Backdoor training generally forces a learned conditional: *if trigger feature fires → shift output distribution toward attacker target*, while preserving clean performance. <sup>1</sup>

In NLP, triggers can be (i) **learned combinations of word substitutions** (harder to notice, more robust than single keywords) or (ii) **style-manipulation triggers** (implicit linguistic style as hidden key). <sup>3</sup>

Prompt- and instruction-based LLM pipelines can also be backdoored (e.g., prompt-as-trigger, instruction-finetuning backdoors), reinforcing that “natural-language space” is a realistic search domain. <sup>4</sup>

Related evaluation mindsets appear in backdoor benchmarks and scanning competitions (e.g., BackdoorBench; NIST <sup>5</sup>’s TrojAI <sup>6</sup>). <sup>7</sup>

## Experimental blueprint for Google Colab <sup>8</sup> black-box batches

This assumes batched prompts against a hosted endpoint (often via Hugging Face <sup>9</sup> in challenge settings run by Jane Street <sup>10</sup>).

### Data design (factorial):

Choose 200–500 benign base prompts (P), spanning multiple task types; generate 1,000–10,000 perturbation instances (T) from pattern families; test each instance on K=10–30 base prompts (replicates) to measure stability.

**Batching:** concurrency 8–32; micro-batches 20–100; exponential backoff on 429/5xx; log all raw prompts/outputs and decode params.

**Per-output features:** output embedding (SBERT-style sentence embeddings), length stats, refusal markers, stylometry, and a crude entropy proxy (compression ratio) as a heuristic. <sup>11</sup>

**Tests + stopping:** use distributional two-sample tests (MMD / energy distance) and clustering diagnostics (silhouette + gap statistic). Apply BH-FDR across hypotheses; stop expanding a family when additional samples stop producing new separable clusters or no candidates survive validation. <sup>12</sup>

```
flowchart TD
  A[Sample base prompts P] --> B[Baseline queries]
  B --> C[Embeddings + stylometry + refusal features]
  C --> D[Generate perturbations by pattern family]
  D --> E[Apply each perturbation to K prompts]
  E --> F[Batched endpoint queries]
  F --> G[Drift + two-sample tests + clustering]
  G --> H{Flagged?}
  H -- yes --> I[Minimality + paraphrase robustness]
  H -- no --> D
  I --> J[Holdout confirmation + reproducibility log]
```

## Structural pattern-class taxonomy to sweep

A backdoor keyed in natural language is commonly detected faster by **family sweeps** than by random string guessing, especially when triggers are multi-token or style-based. <sup>13</sup>

| Class                    | Parameters            | Parametric generation rule                        |
|--------------------------|-----------------------|---------------------------------------------------|
| Prefixes                 | phrase p, separator s | <code>p + s + BASE</code>                         |
| Suffixes                 | phrase p              | <code>BASE + "\n\n" + p</code>                    |
| Wrappers                 | left L, right R       | <code>L + BASE + R</code>                         |
| Positional constraints   | token t, position i   | insert t at token i (start/middle/end)            |
| Repetition thresholds    | phrase p, count n     | repeat p n times (contiguous/spaced)              |
| Formatting               | format f              | markdown quote/code, JSON/YAML shell, lists, etc. |
| Punctuation-heavy        | punct pattern q       | interleave punctuation with phrase                |
| Capitalization           | mode m                | ALLCAPS/lower/Title/camel/snake/etc.              |
| Whitespace               | scheme w              | tabs, blank lines, trailing spaces, hard-wrap     |
| Encoded identifiers      | alphabet A, length ℓ  | pronounceable IDs (e.g., <code>tag-AB12</code> )  |
| Semantic templates       | template τ + slots    | fill benign slots (“For calibration: [X]”)        |
| Multi-token conjunctions | tokens a,b,c          | require co-occurrence within a window             |

| Class                      | Parameters        | Parametric generation rule          |
|----------------------------|-------------------|-------------------------------------|
| Paraphrase-robust variants | paraphraser $\pi$ | paraphrase while preserving meaning |

Full parametric rules **plus 10 safe example instances per class** are provided in the downloadable appendix.

## Measurement, detection, and validation plan

**Embedding-based drift:** compute baseline centroid  $\mu$ ; define drift  $d = 1 - \cos(e, \mu)$ . Backdoor activation should show a stable, replicable drift tail or a distinct cluster across many base prompts.

### Distributional tests:

MMD provides a nonparametric two-sample test in embedding space; energy distance is a complementary distribution metric well-suited to multivariate samples. <sup>14</sup>

### Clustering + separability:

Use silhouette to quantify separation and gap statistic to pick  $k$ ; optionally use HDBSCAN to find stable density clusters and outliers. <sup>15</sup>

### Token-distribution shift without logprobs:

Approximate with your own tokenizer on outputs; compute Jensen–Shannon divergence (more stable than raw KL) as a secondary signal.

### Multiple hypotheses + thresholds:

Apply Benjamini–Hochberg FDR across instances/families; a practical flagging rule is  $q \leq 0.01$  plus effect-size gating (e.g., drift median above 99.5th percentile of sham controls) and replicate stability  $\geq 70\%$ . <sup>16</sup>

### Minimality and robustness:

Confirm candidates by (i) single-character deletion, whitespace/case flips, punctuation swaps; (ii) paraphrasing surrounding context vs paraphrasing the candidate itself; (iii) holdout base prompts and reruns for reproducibility.

## Visualizations, code, and deliverable artifacts

Recommended plots: PCA/UMAP embedding maps, drift time-series across sweep order (change-point signatures), heatmaps (family  $\times$  effect size /  $q$ ), and cluster diagnostics plots. <sup>17</sup>

Colab-ready async request + embedding/drift snippets are included in the appendix, along with a **corpus-driven candidate generator** that outputs a ranked set of language-like phrases for screening rather than a hand-curated “trigger list.” <sup>18</sup>

[Download the full appendix with tables and code](#)

1 **BadNets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain**

[https://arxiv.org/abs/1708.06733?utm\\_source=chatgpt.com](https://arxiv.org/abs/1708.06733?utm_source=chatgpt.com)

2 8 12 14 **jmlr.org**

<https://www.jmlr.org/papers/volume13/gretton12a/gretton12a.pdf>

3 10 13 **Turn the Combination Lock: Learnable Textual Backdoor Attacks via Word Substitution**

[https://arxiv.org/abs/2106.06361?utm\\_source=chatgpt.com](https://arxiv.org/abs/2106.06361?utm_source=chatgpt.com)

4 **PoisonPrompt: Backdoor Attack on Prompt-based Large Language Models**

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6 7 **proceedings.neurips.cc**

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